DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B total parameters · 21B activated · 128K context window · 8.1T tokens pretraining

236B	21B	128K	42.5%	93.3%	5.76×
Total Parameters	Activated / Token	Context Window	Training Cost ↓	KV Cache ↓	Generation Throughput ↑

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

THE PROBLEM

- **Prohibitive Training Cost**

  Dense dense Transformer scaling drives training compute into the billions of GPU-hours. Every doubling demands 4× more FLOPs.

- **KV Cache Bottleneck**

  Full key-value matrices for every token explode GPU memory during autoregressive generation. Context length amplifies this linearly.

- **Slow Inference**

  With 128K context, standard MHA stores K/V for millions of vectors. Memory bandwidth becomes the limiting factor.

- **MoE Trade-offs**

  Naive MoE reduces activated params but traditional attention still caches full K/V — the bottleneck persists.

WHAT IS NEEDED

- **Efficient Attention**

  An attention mechanism that matches MHA quality while compressing KV cache by >90%.

- **Fine-Grained MoE**

  Expert specialisation beyond coarse top-K routing to capture diverse knowledge.

- **Shared Expert Isolation**

  Dedicated shared experts for common knowledge reduce redundant activation.

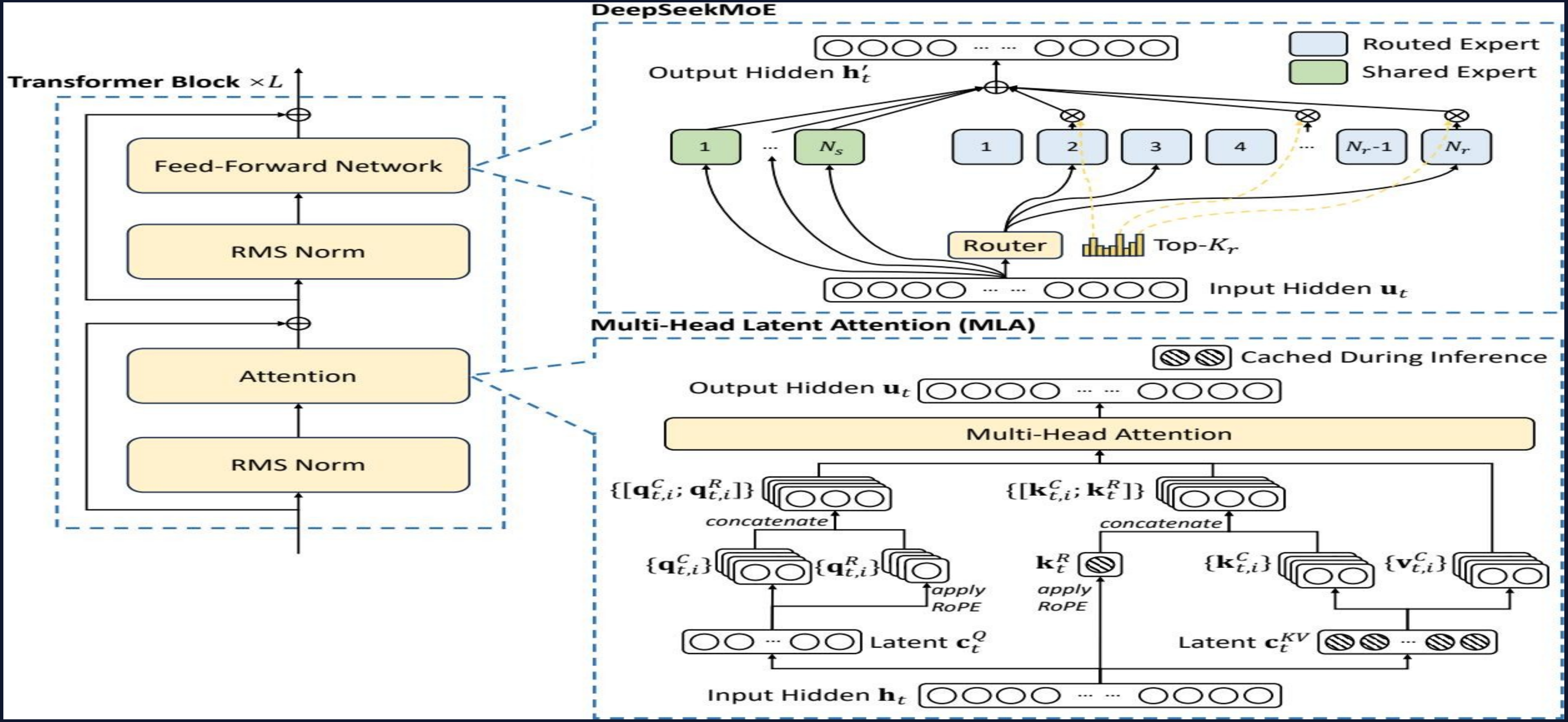
- **Device-Limited Routing**

  Constrain expert assignment to ≤M devices to minimise cross-node communication.

DeepSeek-V2 addresses all four with Multi-Head Latent Attention (MLA) + DeepSeekMoE → **top-tier performance at 21B activated params**

DeepSeek-V2 Architecture

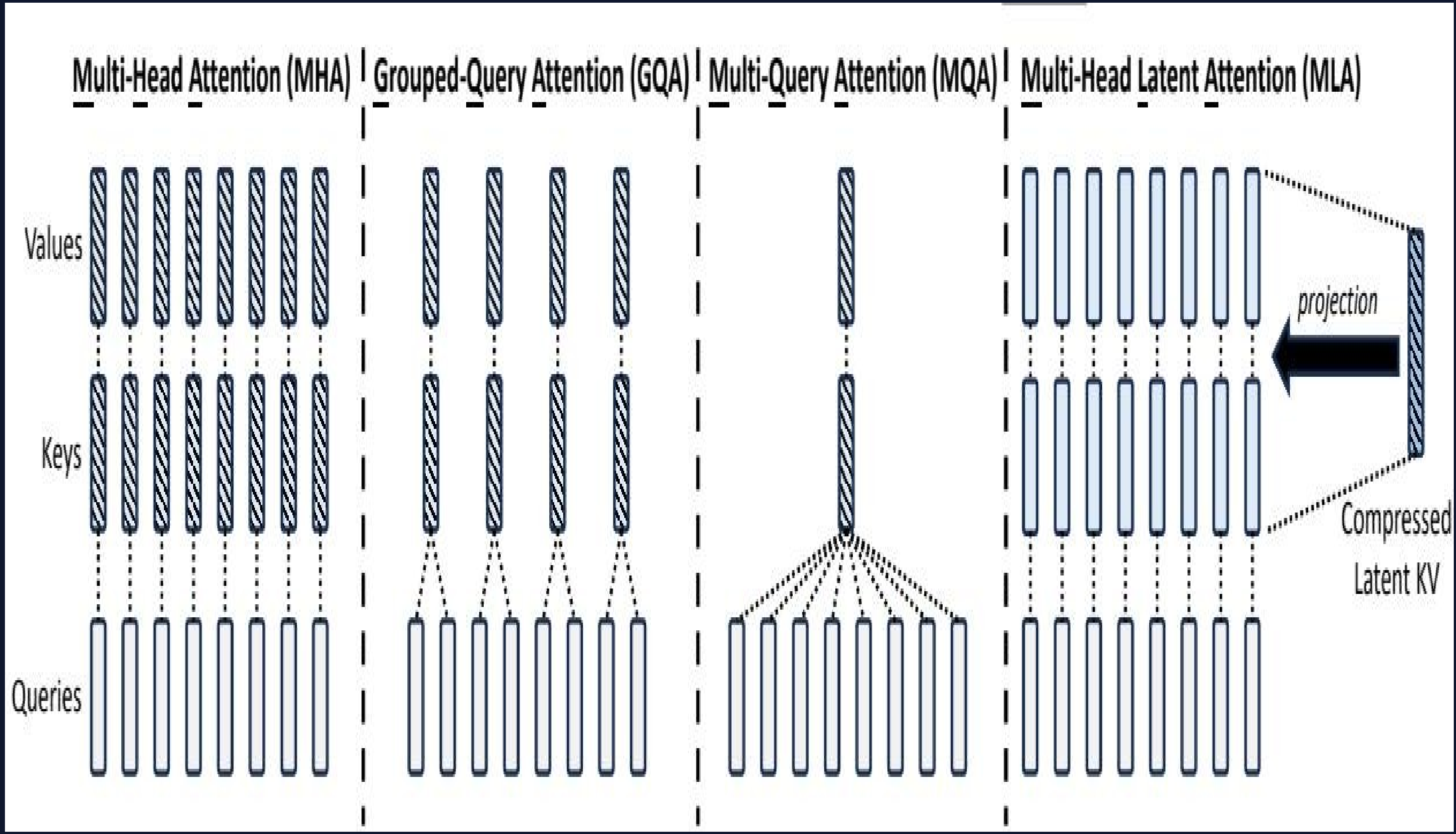
MLA (Multi-Head Latent Attention) + DeepSeekMoE in a Transformer Block



Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

Multi-Head Latent Attention (MLA)

Low-Rank KV Joint Compression Reduces Cache by 93.3% vs. DeepSeek 67B



HOW MLA WORKS

▸ Low-Rank KV Compression

Compress K and V into a shared latent vector c_t^{KV}. Only c_t^{KV} is cached — not full K/V matrices.

▸ Decoupled RoPE

A lightweight decoupled query-side RoPE vector is prepended to the compressed KV for positional information.

▸ Full Reconstruction

Full K and V are recovered on-the-fly at each layer via learned projection matrices — no information loss.

▸ Better than MHA

Compresses KV cache 93.3% while outperforming standard Multi-Head Attention on benchmarks.

▸ vs GQA / MQA

GQA and MQA reduce cache but sacrifice quality. MLA achieves the impossible: better performance, less cache.

Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

DeepSeekMoE Architecture

Fine-Grained Expert Segmentation with Shared Expert Isolation

Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

ARCHITECTURE HIGHLIGHTS

▸ Fine-Grained Expert Segmentation

Split each FFN layer into many small experts. Each token activates only top-K_r of N_r routed experts, increasing specialization granularity.

▸ Shared Expert Isolation

N_s shared experts are always active — they capture common knowledge and world facts without routing decisions, reducing redundancy.

▸ Device-Limited Routing

Each token is routed to experts on ≤ M devices. Reduces cross-node all-to-all communication overhead during distributed training.

▸ Auxiliary Load-Balancing Losses

Three balanced losses (expert-level, device-level, communication-aware) prevent expert collapse and ensure GPU utilisation stays even.

▸ Token-Dropping Strategy (Training)

During training, drop the lowest-priority tokens if expert capacity is exceeded, avoiding overflow without altering the inference graph.

▸ Training Cost Reduction

The combination of sparse activation and communication-efficient routing enables training a 236B model at 42.5% lower cost than DeepSeek 67B dense.

Efficiency Gains vs. DeepSeek 67B

DeepSeek-V2 vs. DeepSeek 67B — Relative Improvements (No Absolute FLOPs / KV Sizes Fabricated)

DeepSeek 67B (Dense)

Architecture:	Dense Transformer — all 67B params activated per token
Attention:	Standard Multi-Head Attention (MHA) — full KV cache per head
Expert MoE:	No MoE — dense FFN layers throughout
Training Cost:	Baseline (reference point for comparison)
KV Cache Size:	Full K/V stored for every token and every head
Generation Throughput:	Baseline (reference point for comparison)

DeepSeek-V2 (MoE + MLA)

Architecture:	MoE Transformer — 236B total / 21B activated per token
Attention:	Multi-Head Latent Attention (MLA) — compressed KV cache
Expert MoE:	DeepSeekMoE: 2 shared + 64 routed experts
Training Cost:	42.5% reduction vs. DeepSeek 67B
KV Cache Size:	93.3% reduction vs. DeepSeek 67B
Generation Throughput:	5.76× improvement vs. DeepSeek 67B

42.5%
Training Cost
Reduction

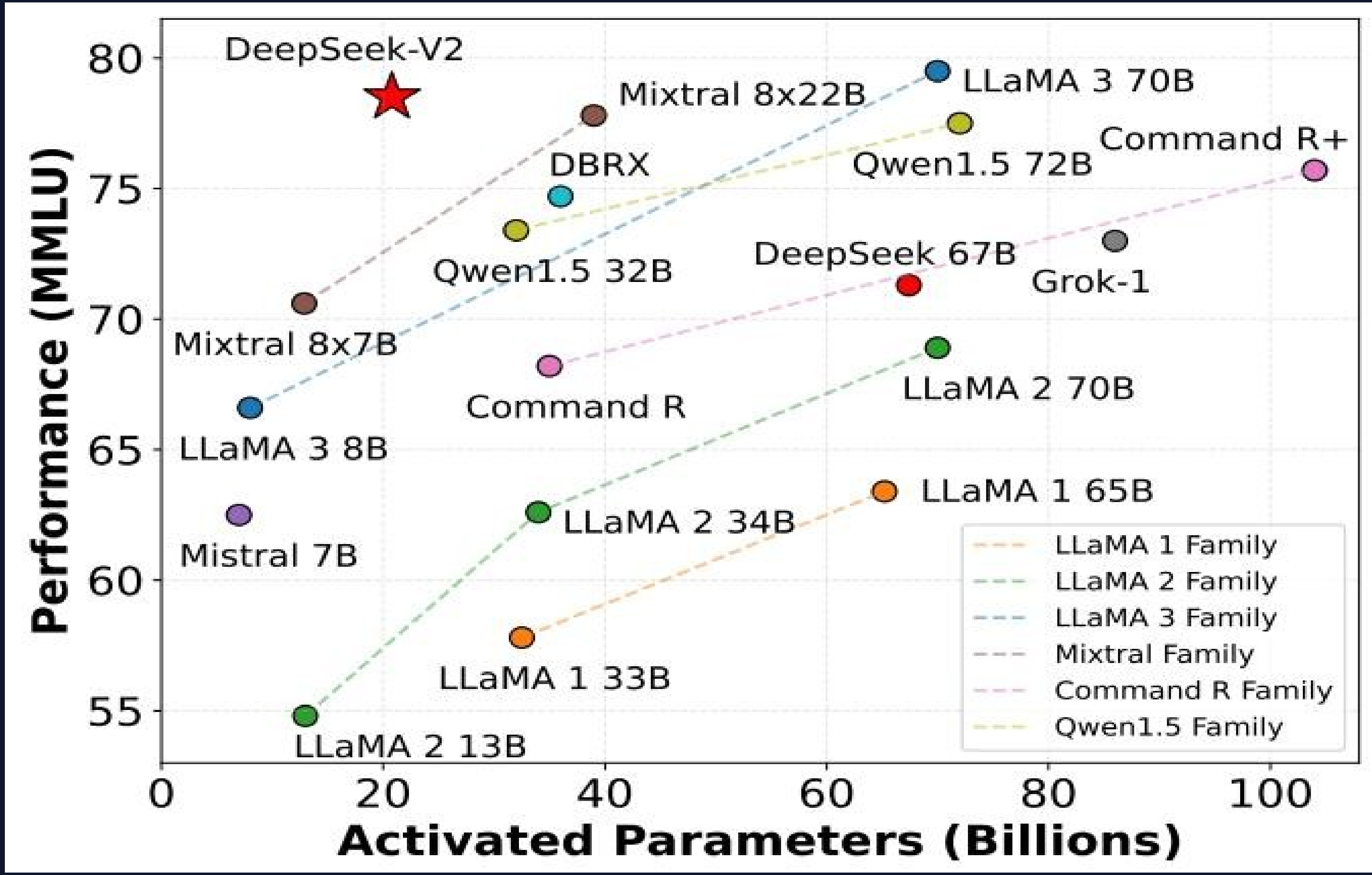
93.3%
KV Cache
Reduction

5.76×
Generation
Throughput ↑

Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

MMLU Benchmark: Pareto-Optimal Efficiency

Only 21B Activated Parameters vs. 70B–72B Competitors — MMLU Score vs. Activated Params



Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

MMLU SCORES

Model	Act. Params	MMLU
DeepSeek-V2	21B	**78.5**
LLaMA 3 70B	70B	77.0
Mixtral 8×22B	39B	77.6
Qwen1.5 72B	72B	75.6
DeepSeek 67B	67B	73.0
LLaMA 2 70B	70B	69.0
Mixtral 8×7B	13B	70.6
LLaMA 3 8B	8B	65.0

★ DeepSeek-V2: top MMLU
at 3.3× fewer activated params

DeepSeek-V2 Chat (RL)

SFT on 1.5M Sessions + GRPO Reinforcement Learning Alignment

Pretrained 236B / 21B → SFT on 1.5M Sessions → GRPO RL Optimization → DeepSeek-V2 Chat (RL)

CHAT EVALUATION RESULTS

38.9%

AlpacaEval 2.0 Length-Controlled Win Rate

Outperforms most open-source and many closed-source models

8.97

MT-Bench Overall Score

Competitive with leading closed-source chatbots

7.91

AlignBench Overall (Chinese)

Best open-source model in Chinese; rivals closed-source systems

Source: DeepSeek-V2 paper (arxiv.org/abs/2405.04434)

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

01

MLA is a Superior KV-Cache Solution

Achieves better performance than standard MHA while compressing the KV cache by 93.3%. A strictly better alternative to GQA/MQA for inference efficiency.

02

DeepSeekMoE Enables Economical Training

Fine-grained experts + shared expert isolation + device-limited routing reduce training costs by 42.5% vs. DeepSeek 67B dense, enabling efficient scaling to 236B parameters.

03

Top-Tier Performance at 21B Activated Params

Matches or exceeds 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU with 3.3× fewer activated parameters — Pareto-optimal efficiency.

04

5.76× Generation Throughput Improvement

Dramatically higher inference throughput vs. DeepSeek 67B makes large-scale deployment practical and cost-effective.

05

Strongest Open-Source Chat Model in Chinese

DeepSeek-V2 Chat (RL) outperforms all open-source and most closed-source models on Chinese benchmarks, with competitive English open-ended evaluation performance.